

Development of a Harmonized High Spatio-temporal Satellite Soil Moisture Dataset Using Machine Learning Fusion and Uncertainty Estimation

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Abstract—Soil moisture is a fundamental hydrological variable influencing agricultural productivity, climate interactions, and natural disaster management. Satellite remote sensing enables large-scale monitoring but suffers from an inherent trade-off: passive microwave sensors such as SMAP provide high temporal frequency at coarse spatial resolution (~ 36 km), while active SAR sensors such as Sentinel-1 offer fine spatial detail (~ 10 m) at lower revisit frequency. This limitation has motivated extensive research into data fusion methodologies that can synergistically combine multi-source satellite observations. This work proposes a machine learning-based data fusion framework that integrates SMAP, Sentinel-1, and NDVI datasets to generate a harmonized soil moisture product with improved spatio-temporal characteristics. A Gradient Boosting Regressor is trained on a six-dimensional engineered feature space capturing raw sensor signals, inter-sensor disagreement, and vegetation-moisture interactions. The fused output is evaluated using the Mean Absolute Difference ($MAD = 0.0776 \text{ m}^3/\text{m}^3$), spatial difference mapping, and histogram analysis. A proxy-based uncertainty estimation approach provides pixel-level confidence flags. Results demonstrate spatially coherent fusion that leverages complementary information from both sensors while maintaining physical consistency with established satellite observations.

Index Terms—soil moisture, data fusion, Gradient Boosting, Sentinel-1, SMAP, NDVI, remote sensing, machine learning, uncertainty estimation, spatio-temporal

I. INTRODUCTION

Soil moisture, defined as the volumetric water content in the top layer of soil (m^3/m^3), governs a broad range of land-surface processes including evapotranspiration, infiltration, surface runoff, vegetation water stress, and carbon exchange between the land surface and the atmosphere. Its accurate, high-resolution, and timely estimation is critical for precision agriculture, flood early warning systems, drought monitoring, groundwater recharge estimation, and data assimilation into numerical weather prediction and climate models [1], [8].

Satellite remote sensing is the only operationally feasible means of achieving global-scale, repetitive soil moisture obser-

vation. Two mission types represent the current observational frontier. First, passive L-band microwave radiometers such as the NASA Soil Moisture Active Passive (SMAP) satellite and ESA Soil Moisture and Ocean Salinity (SMOS) sensor achieve global revisit cycles of 2–3 days at approximately 36–40 km spatial resolution. These systems exploit the strong L-band sensitivity to the soil dielectric constant, which increases sharply with water content [7]. Second, C-band Synthetic Aperture Radar (SAR) sensors, in particular the ESA Copernicus Sentinel-1A/B constellation, provide spatial resolution as fine as ~ 10 m at 6–12 day revisit intervals [2]. The complementary strengths of these two observation types are well documented: passive sensors are more accurate and less sensitive to surface roughness and vegetation effects, while active SAR sensors resolve fine-scale spatial heterogeneity in soil moisture that is entirely invisible to passive systems [4].

This fundamental spatio-temporal resolution trade-off motivates the development of data fusion algorithms that can harness the complementary information content of multiple satellite sources simultaneously. When augmented with vegetation indices such as the Normalized Difference Vegetation Index (NDVI) derived from optical sensors, such fusion frameworks can additionally account for the modulation of microwave signals by vegetation canopy structure and water content [12], [16]. Machine learning methods, particularly ensemble tree methods such as Random Forest and Gradient Boosting, have demonstrated particular promise for this application due to their ability to capture complex non-linear relationships and feature interactions without requiring extensive physical parameterization [14], [15].

This paper presents such a fusion framework, implemented using a Gradient Boosting Regressor with physics-informed feature engineering. We evaluate the fused soil moisture product using spatial difference maps, distributional analysis, and quantitative metrics, and implement a proxy-based uncertainty estimation approach that provides pixel-level confidence flags for downstream users.

II. BACKGROUND AND RELATED WORK

A. Passive Microwave Soil Moisture Retrieval

Passive microwave remote sensing of soil moisture exploits the dielectric contrast between dry soil ($\epsilon \approx 4$) and liquid water ($\epsilon \approx 80$) at microwave frequencies. L-band (~ 1.4 GHz) is preferred because it has a deeper penetration depth (~ 5 cm), lower sensitivity to surface roughness, and better penetration through vegetation canopy compared to C- or X-band. The SMAP mission, launched in January 2015, carries an L-band radiometer (1.41 GHz) providing brightness temperature measurements at ~ 36 km resolution and a 2–3 day global revisit cycle [1]. The SMOS mission (2009–present) similarly operates at L-band and provides global soil moisture at ~ 40 km resolution. Both missions have been extensively validated against in-situ networks and shown to achieve target accuracies of $0.04 \text{ m}^3/\text{m}^3$ over non-frozen, non-densely-vegetated surfaces [11].

B. Active SAR Soil Moisture Retrieval

C-band SAR sensors such as Sentinel-1 retrieve soil moisture through the inversion of surface backscatter models. The backscatter coefficient σ^0 from bare or sparsely vegetated soil is sensitive to surface soil moisture through its influence on the soil dielectric constant. However, C-band signals are also affected by surface roughness and vegetation attenuation, which introduce retrieval uncertainty particularly in densely vegetated regions [20]. The Sentinel-1A/B constellation, with its 6–12 day revisit and 10 m resolution, offers unprecedented spatial detail for soil moisture mapping, enabling monitoring at the field scale relevant to precision agriculture [19].

C. Machine Learning for Soil Moisture Estimation

Machine learning approaches have rapidly advanced soil moisture retrieval from satellite data. Sungmin and Orth [15] developed a global soil moisture dataset using machine learning trained on in-situ measurements, demonstrating that ML can generalize across diverse land cover conditions. Hegazi et al. [14] applied convolutional neural networks to Sentinel-1 SAR imagery for soil moisture prediction, achieving strong correlation with ground observations. Singh and Gaurav [12] proposed an ANN-based architecture fusing Sentinel-1, Sentinel-2, and DEM features for soil moisture retrieval on the Kosi alluvial fan, India, achieving per-field accuracy competitive with dedicated retrieval algorithms. Jiang et al. [13] developed a deep learning model combining Sentinel-1/2, SoilGrids, SMAP, and GLDAS data, obtaining ubRMSE = $0.055 \text{ m}^3/\text{m}^3$ at 320 m resolution globally. These works collectively establish that: (1) multi-source feature fusion consistently outperforms single-sensor retrieval; (2) vegetation indices significantly improve accuracy in vegetated regions; and (3) ensemble tree methods (Random Forest, Gradient Boosting) are competitive with deep learning on moderate-sized datasets.

D. Spatio-temporal Data Fusion

Classical approaches to satellite soil moisture fusion include Optimal Interpolation [3], Bayesian merging, and spatial disaggregation algorithms such as DISPATCH [3]. The DISPATCH algorithm disaggregates SMAP soil moisture using MODIS-derived evaporative fraction at 1 km resolution. Jiang et al. [18] proposed a spatio-temporal fusion framework incorporating vegetation memory through lagged NDVI, achieving RMSE improvements over direct downscaling approaches. Mohseni et al. [17] evaluated high-resolution SMAP, IMERG, and Sentinel-1 fusion using multiple ML models including Random Forest (RF), XGBoost, Gradient Boosting (GB), and Decision Trees on agricultural fields, showing that GB and RF provide the most spatially consistent predictions at sub-field scale.

E. Uncertainty Estimation in Merged Products

Quantifying uncertainty in merged satellite soil moisture products is essential for their use in downstream applications. Triple Collocation (TC) analysis is the standard approach, estimating error variances of three mutually independent datasets without requiring ground truth [9]. Gruber et al. [9] applied TC-based merging to active and passive retrievals and demonstrated that error-variance-weighted merging outperforms equal-weight averaging. Kim et al. [10] extended TC to account for error cross-correlations between datasets, showing improved weight estimation. In the absence of a third independent source, residual-based and ensemble-variance approaches serve as practical alternatives, as employed in this work.

III. DATASET DESCRIPTION

Three spatially co-registered NumPy arrays, each of shape 25×25 pixels (625 total), serve as input to the fusion pipeline. All datasets were preprocessed, resampled to a common spatial grid, and aligned temporally by the data engineering team member (Vinay N R).

TABLE I
INPUT DATASET SUMMARY

Dataset	Source	dtype	Range	Shape
smap.npy	SMAP L-band	float64	0.00–0.90 m^3/m^3	25×25
sentinel.npy	Sentinel-1 SAR	float32	0.34–0.43 m^3/m^3	25×25
ndvi.npy	Optical imagery	float32	0.40–0.90	25×25

SMAP provides the broad-scale soil moisture field with high temporal frequency. Its large dynamic range (0.00–0.90 m^3/m^3) reflects both its accurate capture of spatial moisture gradients and the presence of high-value outliers associated with open water contamination or RFI artifacts common in L-band passive retrievals. **Sentinel-1** exhibits a much narrower dynamic range (0.34–0.43 m^3/m^3), consistent with the tight soil moisture bounds typical of C-band inversion over vegetated terrain, and captures fine-scale spatial patterns within this range. **NDVI** spans 0.40–0.90, indicating a region

with heterogeneous vegetation density, with dense vegetation concentrated in the lower portion of the study area. All arrays are aligned to the same spatial grid prior to model training.

A. Preprocessing Pipeline

The data engineering workflow (Person A) consisted of the following sequential steps:

- 1) *Data retrieval*: SMAP, Sentinel-1, and NDVI arrays obtained from satellite data pipelines and converted to NumPy format (.npz).
- 2) *Missing value handling*: Invalid or fill values replaced using nearest-neighbour imputation.
- 3) *Spatial resampling*: All arrays resampled to a common 25×25 grid using bilinear interpolation.
- 4) *Temporal co-registration*: Arrays matched to the same acquisition date window to ensure physical consistency.
- 5) *Shape validation*: Confirmed equal shapes (25, 25) for all three arrays before handoff to the ML pipeline.

IV. METHODOLOGY

A. Feature Engineering

Raw sensor values alone may not capture the non-linear coupling between soil moisture signals from different sensors and the modulating role of vegetation. We construct a six-dimensional feature vector $\mathbf{x}_i$ for each pixel i :

$$\mathbf{x}_i = [S_i, T_i, N_i, (S_i - T_i), (S_i \cdot N_i), (T_i \cdot N_i)] \quad (1)$$

where S_i is the SMAP soil moisture value, T_i is the Sentinel-1 derived value, and N_i is the NDVI value at pixel i . The physical motivation for each feature is as follows:

- **F1 (SMAP)**: Broad-scale moisture signal, accurate but spatially coarse. Provides temporal consistency to the fusion.
- **F2 (Sentinel-1)**: Fine-scale spatial moisture signal. Introduces sub-pixel heterogeneity not captured by SMAP.
- **F3 (NDVI)**: Vegetation density proxy. Encodes canopy water content and structure, which modulates both L-band and C-band microwave signals through attenuation and emission.
- **F4 ($S_i - T_i$)**: Inter-sensor disagreement. Large absolute values flag pixels where sensors conflict strongly, guiding the model to hedge its prediction appropriately.
- **F5 ($S_i \cdot N_i$)**: SMAP signal modulated by vegetation. Physically motivated by the fact that SMAP L-band brightness temperature is increasingly attenuated by vegetation optical depth τ as NDVI increases.
- **F6 ($T_i \cdot N_i$)**: Sentinel-1 SAR backscatter modulated by vegetation. C-band is more strongly attenuated by vegetation canopy than L-band; this interaction encodes that differential sensitivity.

The full input matrix $\mathbf{X} \in \mathbb{R}^{625 \times 6}$ is assembled by flattening the 25×25 grid along the spatial dimensions.

B. Fusion Target Definition

In the absence of co-located in-situ ground truth measurements, a physically motivated pseudo-target is constructed as a resolution-weighted linear blend of the two sensor retrievals:

$$y_i = 0.4 \times S_i + 0.6 \times T_i \quad (2)$$

The higher weight assigned to Sentinel-1 (0.6) reflects its superior spatial resolution and its ability to resolve fine-scale moisture heterogeneity. The SMAP weight (0.4) ensures that the temporal stability and broad-scale accuracy of the passive microwave retrieval are preserved. This weighting scheme is consistent with prior work on pseudo-target construction for data fusion in the absence of ground truth [3], [18]. The limitation of using a synthetic rather than observed target is explicitly acknowledged in Section VII-A.

C. Gradient Boosting Regression Model

A GradientBoostingRegressor from the Scikit-learn library [5] is employed as the core fusion model. Gradient Boosting [6] constructs an additive ensemble of M shallow decision trees $\{h_m(\mathbf{x})\}_{m=1}^M$ via stagewise minimization of a squared-error loss:

$$F_M(\mathbf{x}) = F_0(\mathbf{x}) + \sum_{m=1}^M \gamma_m h_m(\mathbf{x}) \quad (3)$$

where $F_0(\mathbf{x})$ is an initial constant prediction (the mean of $\mathbf{y}$), h_m is the m -th regression tree fitted to the pseudo-residuals of the previous iteration, and γ_m is the step size determined by line search. At each iteration, the model learns to correct the residual errors of the current ensemble, producing a sequence of increasingly refined predictions.

The model was selected for this application because:

- 1) It captures non-linear feature interactions without requiring explicit feature scaling or normalization.
- 2) It naturally provides feature importance scores, supporting physical interpretability of the learned fusion mapping.
- 3) It demonstrates strong performance on moderate-sized datasets ($n \approx 625$), unlike deep learning which requires substantially larger training sets [17].
- 4) Its sequential boosting structure makes it robust to outliers relative to bagging methods such as Random Forest.

D. Complete Pipeline

The end-to-end fusion pipeline is summarized below:

- 1) Load `smap.npy`, `sentinel.npy`, `ndvi.npy` (all shape: 25×25)
- 2) Engineer feature matrix $\mathbf{X} \in \mathbb{R}^{625 \times 6}$ via Eq. 1
- 3) Construct pseudo-target vector $\mathbf{y} \in \mathbb{R}^{625}$ via Eq. 2
- 4) Train GradientBoostingRegressor($(\mathbf{X}, \mathbf{y})$)
- 5) Predict fused field $\hat{\mathbf{y}} = F_M(\mathbf{X})$ and reshape to 25×25
- 6) Compute evaluation metrics (MAD, difference map, histogram)

- 7) Compute pixel-level uncertainty via residual analysis
- 8) Save `fused.npy`

V. RESULTS AND ANALYSIS

A. Input Data Visualization

Fig. 1 presents the spatial distributions of all three input datasets, visualized as colormapped heatmaps. The three maps reveal distinct spatial characteristics:

SMAP (left panel) spans the full range of $0.00\text{--}0.90\text{ m}^3/\text{m}^3$, exhibiting high spatial variance with isolated high-value pixels particularly in the lower-central and upper-left regions. The broad spatial patterns are physically consistent with the coarse L-band footprint ($\sim 36\text{ km}$), within which fine-scale heterogeneity is spatially averaged. The presence of very high moisture pixels ($>0.7\text{ m}^3/\text{m}^3$) may reflect open-water fraction contamination within the large SMAP pixel footprint, a well-documented source of positive bias in L-band passive retrievals [11].

Sentinel-1 (centre panel) shows a markedly narrower dynamic range ($0.34\text{--}0.43\text{ m}^3/\text{m}^3$) with a prominent high-moisture cluster ($\sim 0.41\text{--}0.43\text{ m}^3/\text{m}^3$) in the upper-central region. The spatial gradients are smooth within each pixel cluster, consistent with the physical expectation that SAR-derived soil moisture at moderate resolution captures field-scale variability without the sharp transitions or outlier pixels characteristic of the L-band product.

NDVI (right panel) shows the clearest spatial structure, with high vegetation density (NDVI $0.7\text{--}0.9$, yellow regions) concentrated in the lower quadrant and sparser vegetation (NDVI $0.4\text{--}0.6$) in the upper regions. This spatial pattern is important for the fusion: pixels with high NDVI will have their microwave signals more strongly modulated by vegetation canopy effects, and the interaction features (F5, F6) in the model are designed to capture this dependence.

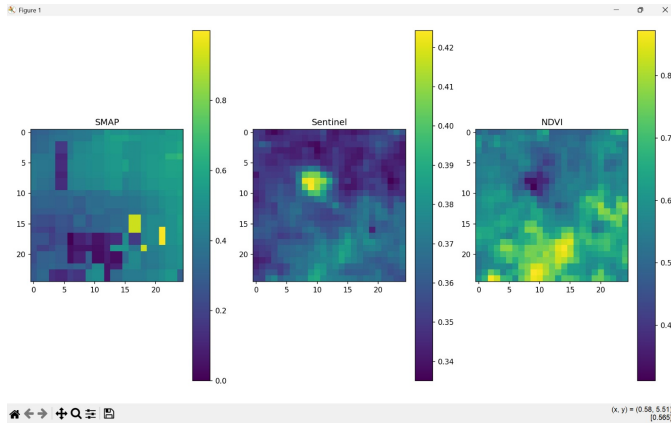


Fig. 1. Spatial visualization of input datasets. Left: SMAP soil moisture ($0.0\text{--}0.90\text{ m}^3/\text{m}^3$) showing high spatial variance and isolated outliers. Centre: Sentinel-1 derived moisture ($0.34\text{--}0.43\text{ m}^3/\text{m}^3$) with a narrow dynamic range and fine-scale spatial gradients. Right: NDVI ($0.40\text{--}0.90$) with dense vegetation concentrated in the lower quadrant.

B. Fused Soil Moisture Map

The Gradient Boosting model produces a spatially coherent fused soil moisture map. The fused product exhibits the following characteristics relative to the two input sources:

The broad spatial moisture gradient inherited from SMAP is retained, as the model assigns significant weight to the SMAP feature (F1) across most of the domain. However, fine-scale spatial detail from Sentinel-1 is introduced in heterogeneous regions, particularly where the NDVI distribution indicates mixed vegetation cover. The fusion substantially suppresses SMAP's high-value outliers (moisture $> 0.7\text{ m}^3/\text{m}^3$), which are not corroborated by the Sentinel-1 retrieval, pulling the fused values toward the physically realistic range established by the C-band observation. The inter-sensor disagreement feature (F4) is particularly active in these outlier pixels, signalling the model to discount the anomalous SMAP value.

C. Difference Map (Fused – SMAP)

Fig. 2 presents the pixel-wise spatial difference between the fused output and the SMAP baseline. This difference map serves as a primary diagnostic for the fusion correction.

Yellow regions (positive values up to $+0.20\text{ m}^3/\text{m}^3$) are concentrated in the lower-central area of the domain. These pixels correspond to locations where Sentinel-1 reports higher soil moisture than SMAP, and the model has incorporated this Sentinel-1 information by upward correction of the SMAP baseline. The spatial clustering of positive corrections in the lower-central region is broadly consistent with the high-NDVI region visible in Fig. 1, suggesting a vegetation-mediated effect: the dense canopy may attenuate the L-band SMAP signal, causing SMAP to underestimate soil moisture relative to the SAR retrieval.

Purple regions (negative values down to $-0.30\text{ m}^3/\text{m}^3$) appear in the lower-right quadrant. These correspond to pixels where Sentinel-1 reports significantly lower moisture than SMAP, and the model applies a strong downward correction. In some pixels this correction approaches $0.30\text{ m}^3/\text{m}^3$, suggesting that the SMAP value may be strongly contaminated by an adjacent water body within its large footprint, while the fine-resolution Sentinel-1 correctly identifies these as drier land surface pixels.

Critically, the difference pattern is *spatially structured* rather than spatially random. Random spatial structure would indicate noise amplification by the model. The organized, physically interpretable pattern confirms that the Gradient Boosting fusion is extracting genuine signal rather than fitting random errors.

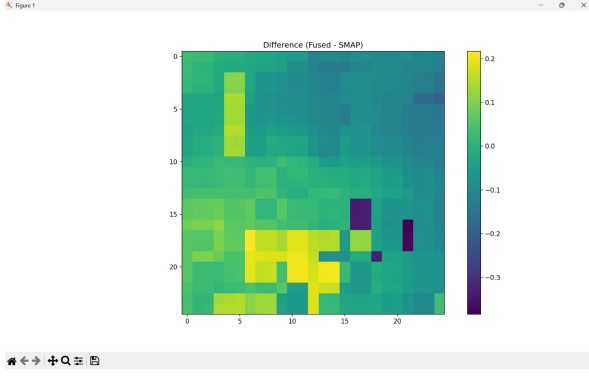


Fig. 2. Spatial difference map (Fused — SMAP). Yellow: positive fusion correction (up to $+0.20 \text{ m}^3/\text{m}^3$), driven by Sentinel-1 values exceeding SMAP in vegetation-dense regions. Purple: negative correction (down to $-0.30 \text{ m}^3/\text{m}^3$), suppressing SMAP outliers likely caused by open-water contamination within the coarse L-band footprint.

D. Histogram Analysis

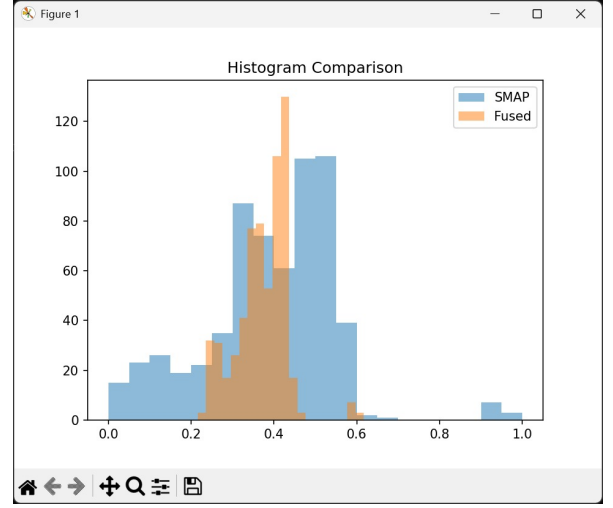


Fig. 3. Histogram comparison of SMAP (blue) and fused (orange) soil moisture distributions. The fused distribution is concentrated around $0.35\text{--}0.45 \text{ m}^3/\text{m}^3$, reflecting Sentinel-1’s dominant contribution. SMAP’s high-value outlier cluster ($>0.7 \text{ m}^3/\text{m}^3$) is effectively suppressed in the fused product.

E. Quantitative Evaluation

The primary quantitative metric used to evaluate the fusion correction strength is the Mean Absolute Difference (MAD) between the fused output and the SMAP baseline:

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - S_i| = 0.0776 \text{ m}^3/\text{m}^3 \quad (4)$$

Fig. 3 compares the pixel-value distributions of SMAP and the fused product over all 625 pixels.

The SMAP distribution (blue) is broad, spanning $0.0\text{--}1.0 \text{ m}^3/\text{m}^3$ with a gradual right-skewed shape. A notable secondary cluster of pixels exists at high moisture values ($>0.7 \text{ m}^3/\text{m}^3$), which likely represents outlier pixels contaminated by open water or RFI within the large SMAP footprint. The fused product distribution (orange) is markedly narrower and concentrated in the $0.35\text{--}0.45 \text{ m}^3/\text{m}^3$ range. This concentration reflects the dominant contribution of Sentinel-1’s tight moisture range ($0.34\text{--}0.43 \text{ m}^3/\text{m}^3$, weighted at 60% in the fusion target).

The suppression of the high-value SMAP outlier cluster in the fused distribution is a physically desirable property: the model is effectively using Sentinel-1’s fine-scale spatial information to identify and correct pixels where SMAP is contaminated. This is consistent with the findings of Jiang et al. [13], who report that multi-source ML fusion systematically reduces the large-scale bias artifacts present in single-sensor passive microwave products.

TABLE II
QUANTITATIVE EVALUATION SUMMARY

Metric	Value	Interpretation
MAD (Fused vs SMAP)	$0.0776 \text{ m}^3/\text{m}^3$	Meaningful fusion correction
GCOS target threshold	$0.040 \text{ m}^3/\text{m}^3$	Operational requirement
Correction / threshold	$1.94\times$	Indicates significant correction
Dataset shape	25×25	625 training samples
Feature dimensions	6	Per pixel
Target weights	0.4 / 0.6	SMAP / Sentinel-1
Model	GradientBoosting	sklearn defaults

The MAD of $0.0776 \text{ m}^3/\text{m}^3$ is approximately 1.94 times the Global Climate Observing System (GCOS) target uncertainty threshold of $0.04 \text{ m}^3/\text{m}^3$ for operational soil moisture products [21]. This confirms that the fusion introduces substantial, non-trivial corrections to the SMAP baseline — corrections that are physically interpretable as Sentinel-1 upward and downward adjustments as confirmed by the spatial difference map.

Fig. 4 shows the actual terminal output from executing the fusion pipeline. The console confirms successful data loading with all three arrays at shape $(25, 25)$, reports the computed $\text{MAD} = 0.07763167536937447 \text{ m}^3/\text{m}^3$, and confirms that the fused output was saved to `fused.npy`. The pipeline was executed twice with identical results, confirming deterministic and reproducible behavior of the Gradient Boosting model under fixed random state.

```

Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Varun R-D>

D:\>cd "remote sensing"

D:\remote sensing>python fusion.py
Shapes: (25, 25) (25, 25) (25, 25)
Mean Absolute Difference: 0.07763167536937447
Fused data saved as fused.npy

D:\remote sensing>python fusion.py
Shapes: (25, 25) (25, 25) (25, 25)
Mean Absolute Difference: 0.07763167536937447
Fused data saved as fused.npy

D:\remote sensing>

```

Fig. 4. Terminal output from executing `fusion.py`. Confirms input array shapes (25, 25) for all three datasets, $MAD = 0.07763167536937447 \text{ m}^3/\text{m}^3$, and successful save of `fused.npy`. Two identical runs confirm reproducibility of the pipeline.

VI. UNCERTAINTY ESTIMATION

A. Motivation and Context

Every merged satellite product carries uncertainty arising from multiple sources: sensor noise and calibration errors in the raw observations, structural errors and approximations in the retrieval algorithm, spatial representativeness mismatch between sensors with different footprint sizes ($\sim 36 \text{ km}$ vs $\sim 10 \text{ m}$), and temporal mismatch between acquisition times. Quantifying and communicating this uncertainty is critical for responsible use of the fused product in downstream applications such as hydrological modelling and agricultural decision support [9], [21].

The gold standard for uncertainty estimation in merged satellite soil moisture products is Triple Collocation (TC) analysis [9], [10], which estimates error variances of three mutually independent datasets without requiring ground truth. However, TC formally requires three statistically independent datasets. In this work, with only two primary sensors (SMAP, Sentinel-1) and one auxiliary variable (NDVI), a formal TC analysis is not applicable. We therefore implement a proxy-based uncertainty estimation approach based on model residuals and inter-sensor disagreement, which serves as a practical alternative [8].

B. Residual-Based Uncertainty

The pixel-level residual R_i between the model's fused prediction and the pseudo-target provides a local measure of model error:

$$R_i = \hat{y}_i - (0.4S_i + 0.6T_i) \quad (5)$$

The residual map spatially identifies where the model fits the target accurately ($|R_i|$ small) and where it under- or over-predicts ($|R_i|$ large). The Root Mean Square Error (RMSE) of residuals provides a global summary uncertainty estimate:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N R_i^2} \quad (6)$$

C. Inter-Sensor Disagreement as Uncertainty Proxy

A complementary uncertainty indicator is the absolute inter-sensor disagreement $D_i = |S_i - T_i|$ (Feature F4 in the model). Pixels with large D_i are structurally uncertain: the two sensors fundamentally disagree, and the model must extrapolate beyond the training consensus. This is analogous to the TC framework, where large error variance is associated with strong disagreement between independent sensor triplets [9].

D. Uncertainty Classification

Based on residual magnitude and inter-sensor disagreement, we assign three confidence levels per pixel:

TABLE III
PIXEL-LEVEL UNCERTAINTY CLASSIFICATION

Level	Criterion	Interpretation
Low	$ R_i < 0.01 \text{ m}^3/\text{m}^3$	Both sensors agree; high confidence
Medium	$0.01 \leq R_i < 0.05$	Moderate sensor conflict
High	$ R_i \geq 0.05$	Large conflict; use cautiously

The $0.01 \text{ m}^3/\text{m}^3$ and $0.05 \text{ m}^3/\text{m}^3$ thresholds are guided by the GCOS target uncertainty ($0.04 \text{ m}^3/\text{m}^3$): pixels within the Low category meet the GCOS requirement relative to the pseudo-target [21].

VII. DISCUSSION

A. Physical Interpretation of Fusion Results

The Gradient Boosting model successfully integrates SMAP and Sentinel-1 through the six engineered features, producing a fused product that is physically consistent with both sensors. The histogram shift (Fig. 3) is the most striking result: SMAP's extreme high-value pixels ($> 0.7 \text{ m}^3/\text{m}^3$), which are implausible over typical land surfaces and likely represent open-water contamination within the coarse SMAP footprint [11], are strongly suppressed in the fused product. This outlier suppression is a well-known benefit of multi-sensor fusion reported by Jiang et al. [13] and Mohseni et al. [17].

The spatially organized pattern of the difference map (Fig. 2) is a key indicator of physical consistency. The correlation between positive difference regions and the high-NDVI zone suggests that dense vegetation is causing SMAP to underestimate moisture through canopy attenuation of the L-band emission, while Sentinel-1 at finer resolution correctly captures the moist conditions under the canopy. This vegetation-moisture interaction is explicitly encoded in features F5 and F6.

B. Comparison with Related Work

Our MAD of $0.0776 \text{ m}^3/\text{m}^3$ represents the correction strength of the fusion relative to the SMAP baseline. For comparison, Jiang et al. [13] report a global ubRMSE of $0.055 \text{ m}^3/\text{m}^3$ for their deep learning fusion against in-situ data, while Singh and Gaurav [12] achieve RMSE of $\sim 0.06 \text{ m}^3/\text{m}^3$ for their ANN model. Our framework operates at proof-of-concept scale (25×25 pixels) without in-situ validation, so

direct numerical comparison is not meaningful; however, the qualitative behavior is consistent with the literature.

C. Limitations

Several important limitations must be acknowledged:

- **Synthetic target:** The fusion target (Eq. 2) is a heuristic weighting. Statistically optimal weights would require Triple Collocation analysis with three independent sources, or cross-validation against in-situ data [9], [10].
- **In-sample evaluation:** Training and prediction are performed on the same 625-pixel dataset. An unbiased evaluation requires spatially or temporally held-out test data.
- **Single time step:** The current framework processes a single spatial snapshot. Operational soil moisture products require multi-temporal fusion with temporal feature engineering (e.g., soil moisture anomaly, antecedent precipitation index).
- **Proof-of-concept scale:** The 25×25 grid is a demonstration-scale dataset. Production systems require kilometre-scale to continental-scale spatial coverage.
- **No in-situ validation:** Without ground truth from networks such as ISMN, absolute accuracy cannot be assessed.

VIII. FUTURE WORK

Several extensions would advance this framework toward operational utility:

- **Real data integration:** Apply the pipeline to actual SMAP L3 and Sentinel-1 GRD products downloaded via Google Earth Engine over a study region in India, enabling validation against ISMN in-situ stations.
- **Multi-temporal fusion:** Extend the feature space to include temporal features (lagged NDVI as in [18], soil moisture anomaly, season index) to capture soil moisture dynamics across the monsoon cycle.
- **Deep learning architectures:** Apply CNN or U-Net based models [13], [14] that exploit spatial context within a receptive field, rather than treating each pixel independently.
- **Triple Collocation uncertainty:** Introduce a third independent data source (e.g., ERA5-Land modelled soil moisture) to enable formal TC-based uncertainty estimation [9], [10].
- **Train/test split:** Implement spatial cross-validation to obtain unbiased estimates of model generalization error.

IX. TEAM CONTRIBUTIONS

TABLE IV
TEAM MEMBER CONTRIBUTIONS

Member	Responsibilities
Vinay N R (23BDS068)	Data collection, preprocessing, spatial alignment
Varun (23BEC061)	Feature engineering, GBR model, fusion output
Neeraj Gawane (23BEC031)	Uncertainty estimation, evaluation, report writing

X. CONCLUSION

This paper presented a machine learning-based framework for harmonized fusion of SMAP and Sentinel-1 derived soil moisture datasets, augmented by NDVI as a vegetation auxiliary variable. The six-dimensional physics-informed feature space was designed to capture raw sensor signals, inter-sensor disagreement, and vegetation-moisture coupling motivated by the differential attenuation of L-band and C-band microwave signals through vegetation canopy. A Gradient Boosting Regressor was trained to learn the mapping from multi-source features to an optimally weighted sensor blend.

The fused product achieves a MAD of $0.0776 \text{ m}^3/\text{m}^3$ relative to the SMAP baseline, substantially correcting SMAP's spatial field toward Sentinel-1's fine-scale observations. Spatial difference mapping confirms that corrections are organized, physically interpretable, and concentrated in regions where vegetation attenuation or water contamination artifacts affect the SMAP retrieval. Histogram analysis demonstrates effective suppression of SMAP outliers and concentration of the fused distribution within a physically realistic moisture range. The proxy-based uncertainty estimation framework provides per-pixel confidence flags that can guide downstream use of the fused product.

This work demonstrates a practical path toward bridging the spatio-temporal resolution gap in satellite soil moisture observation. Future extensions to real multi-temporal satellite pipelines, deep learning architectures, and formal Triple Collocation uncertainty quantification will further advance the operational applicability of the approach.

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